Analog Circuits Project

Parameter Digit Mapping (A - I)

A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**
4	1	9	2	4	2	2	7	0

Design Specifications (Calculated from ID)

Parameter	**Formula / Condition**	**Target Value**
Gain (A)	$32 + 2 \cdot D = 32 + 2(2)$	**36 dB**
Cutoff f_1 (f_L)	$(G + H) = (2 + 7)$	**9 kHz**
Cutoff f_2 (f_H)	$(G + H) \cdot 100 = (2 + 7) \cdot 100$	**900 kHz**
Slopes (M_1, M_2)	$C = 9$ (Odd)	$\mathbf{M_1 = 20, M_2 = -40}$
Output Res. (R_{out})	$F = 2$ (Even) $\rightarrow 3 + F$	**5 kΩ**

January 30, 2026

1 Circuit Description

The circuit contains three stages; all three stages are double-input, double-output differential amplifiers. The first and last stages function as the interface of the total amplifier, ensuring proper input handling and output driving capability. The second stage acts as an intermediate gain stage, utilized to increase the total gain of the amplifier to the required specification.

Transistor Parameters: All stages use NMOS transistors with the following parameters:

$$k = 1\,\frac{mA}{V^2}, \quad \lambda = 0, \quad V_{th} = 0.5V$$

Biasing Strategy: Three current mirrors are implemented, one for each amplifier stage ($M_{9,10}$, $M_{7,8}$, and $M_{3,4}$). These current mirrors act as ideal current sources feeding the differential pairs. Due to the symmetric design of the amplifiers, the tail current splits into equal halves, ensuring that the drain current (I_D) is identical for both transistors in a differential pair.

Frequency Response Implementation:

- **Input DC Blocking:** We utilized capacitor **C8** at the input (V_{in1}) to block the DC component of the source while passing the AC signal.
- **Low Frequency ($M_1 = +20\,dB/dec$):** To achieve the $+20\,dB/dec$ slope at low frequencies, we utilized the coupling capacitors **C2 and C3** between the second and third stages. These capacitors effectively provide a zero at the origin (blocking DC bias propagation) and a pole that determines the lower cutoff frequency (f_L).
- **High Frequency ($M_2 = -40\,dB/dec$):** The specification requires a steep roll-off of $-40\,dB/dec$ at high frequencies, which implies the presence of two dominant poles.
  1. The first pole is created by capacitors **C7 and C6** at the input of the third stage (in parallel with the load resistors of the previous stage), which interact with the output resistance of the second stage.
  2. The second pole is introduced by the differential capacitor **C4**, which bridges the two outputs of the third stage (V_{out1} and V_{out2}). This creates the second pole required to double the roll-off rate to $-40\,dB/dec$ beyond the upper cutoff frequency (f_H).

2 Bias Point Analysis

2.1 First Stage

Starting with the current mirror of the first stage. We identify that **M10** is the diode-connected transistor connected to the reference resistor R_{15}.

Reference Current Calculation (M_{10}): The KVL equation for the reference branch is:

$$I_{D10} = \frac{V_{DD} - V_{GS10}}{R_{15}}$$

Since M_{10} is diode-connected and in saturation:

$$I_{D10} = k(V_{GS10} - V_{TH})^2$$

Equating them:

$$\begin{aligned}
\frac{5 - V_{GS10}}{R_{15}} &= k(V_{GS10} - 0.5)^2 \\
\frac{5 - V_{GS10}}{50k} &= 1m(V_{GS10}^2 - V_{GS10} + 0.25) \\
5 - V_{GS10} &= 50(V_{GS10}^2 - V_{GS10} + 0.25) \\
0 &= 50V_{GS10}^2 - 49V_{GS10} + 7.5
\end{aligned}$$

Solving for V_{GS10}:

$$V_{GS10} = 0.79V \quad (\text{or } 0.19V \text{ which is } < V_{TH})$$

Calculating the current:

$$I_{D10} = \frac{5 - 0.79}{50k} = 84.2\,\mu A$$

Current Mirror (M_9): Since M_9 forms a current mirror with M_{10} and has the same size:

$$I_{D9} = I_{D10} = 84.2\,\mu A$$

M_9 acts as the tail current source for the differential pair.

Differential Pair (M_{11}, M_{12}): Due to symmetry, the tail current I_{D9} splits equally between M_{11} and M_{12}:

$$I_{D12} = I_{D11} = \frac{I_{D9}}{2} = 42.1\,\mu A$$

Gate, Source, and V_{GS} Voltages: First, we calculate the gate voltage V_{G12} via the voltage divider (R_{17}, R_{16}):

$$V_{G12} = V_{DD} \cdot \frac{R_{17}}{R_{17} + R_{16}} = 5 \cdot \frac{8}{18} = 2.22V$$

Next, we find the source voltage V_{S12} using the current equation:

$$\begin{aligned}
I_{D12} &= \frac{k}{2}(V_{GS12} - V_{TH})^2 \\
V_{S12} &= V_{G12} - V_{TH} \pm \sqrt{\frac{I_{D12}}{k/2}} \\
V_{S12} &= 2.22 - 0.5 \pm \sqrt{\frac{42.1\mu}{500\mu}} \\
V_{S12} &= 1.51V \quad (\text{Selected to keep } V_{GS} > V_{TH})
\end{aligned}$$

With V_{S12} determined, we calculate the Gate-Source voltage:

$$V_{GS12} = V_{G12} - V_{S12} = 2.22V - 1.51V = 0.71V$$

This confirms the transistor is conducting since $V_{GS12}(0.71V) > V_{TH}(0.5V)$.
Validation of Saturation (M_{12}): To validate saturation, we calculate the drain voltage V_{D12} using the Thevenin equivalent of the load resistors (R_2, R_3):

$$R_{th,D12} = R_2 \parallel R_3 = 4.25\,k\Omega$$

$$V_{th,D12} = V_{DD} \cdot \frac{R_3}{R_3 + R_2} = 5 \cdot \frac{8.5}{17} = 2.5V$$

$$V_{D12} = V_{th,D12} - I_{D12} \cdot R_{th,D12} = 2.5 - (42.1\mu \cdot 4.25k) = 2.32V$$

Finally, we check the saturation condition $V_D > V_G - V_{TH}$:

$$2.32V > 2.22V - 0.5V \implies 2.32V > 1.72V$$

The assumption is correct; M_{12} is in saturation.

2.2 Second Stage

Starting with the current mirror where **M8** is the diode-connected reference transistor (resistor R_1).
Reference Current Calculation (M_8):

$$\frac{5 - V_{GS8}}{R_1} = k(V_{GS8} - V_{TH})^2$$

$$\frac{5 - V_{GS8}}{10k} = 1m(V_{GS8} - 0.5)^2$$

$$0 = 10V_{GS8}^2 - 9V_{GS8} - 2.5$$

Solving gives $V_{GS8} = 1.12V$.

$$I_{D8} = \frac{5 - 1.12}{10k} = 388\,\mu A$$

Current Mirror (M_7):

$$I_{D7} = I_{D8} = 388\,\mu A$$

Differential Pair (M_5, M_6):

$$I_{D5} = \frac{I_{D7}}{2} = 194\,\mu A$$

Gate voltage is set by the previous stage: $V_{G5} = V_{D12} = 2.32V$. Calculate Source Voltage:

$$V_{S5} = V_{G5} - V_{TH} - \sqrt{\frac{I_{D5}}{k/2}} = 1.38V$$

Drain Voltage:

$$V_{D5} = V_{DD} - I_{D5} \cdot R_{13} = 2.2V$$

Saturation Check: $2.2 > 2.32 - 0.5$ (Confirmed).

2.3 Third Stage

Starting with the current mirror where **M4** is the diode-connected reference transistor (resistor R_4).

Reference Current Calculation (M_4):

$$\frac{5 - V_{GS4}}{R_4} = k(V_{GS4} - V_{TH})^2$$

$$\frac{5 - V_{GS4}}{5k} = 1m(V_{GS4} - 0.5)^2$$

$$0 = 5V_{GS4}^2 - 4V_{GS4} - 3.75$$

Solving gives $V_{GS4} = 1.35V$.

$$I_{D4} = \frac{5 - 1.35}{5k} = 730\,\mu A$$

Current Mirror (M_3):

$$I_{D3} = I_{D4} = 730\,\mu A$$

Differential Pair (M_1, M_2):

$$I_{D1} = \frac{I_{D3}}{2} = 365\,\mu A$$

Gate voltage (R_{11}, R_{12} divider): $V_{G1} = 2.5V$. Calculate Source Voltage:

$$V_{S1} = V_{G1} - V_{TH} - \sqrt{\frac{I_{D1}}{k/2}} = 1.4V$$

Drain Voltage:

$$V_{D1} = V_{DD} - I_{D1} \cdot R_5 = 4.09V$$

Saturation Check: $4.09 > 2.5 - 0.5$ (Confirmed).

2.4 Bias Point Verification

The calculated bias points were verified via simulation. Figure 1 shows the DC operating points matching the theoretical analysis.

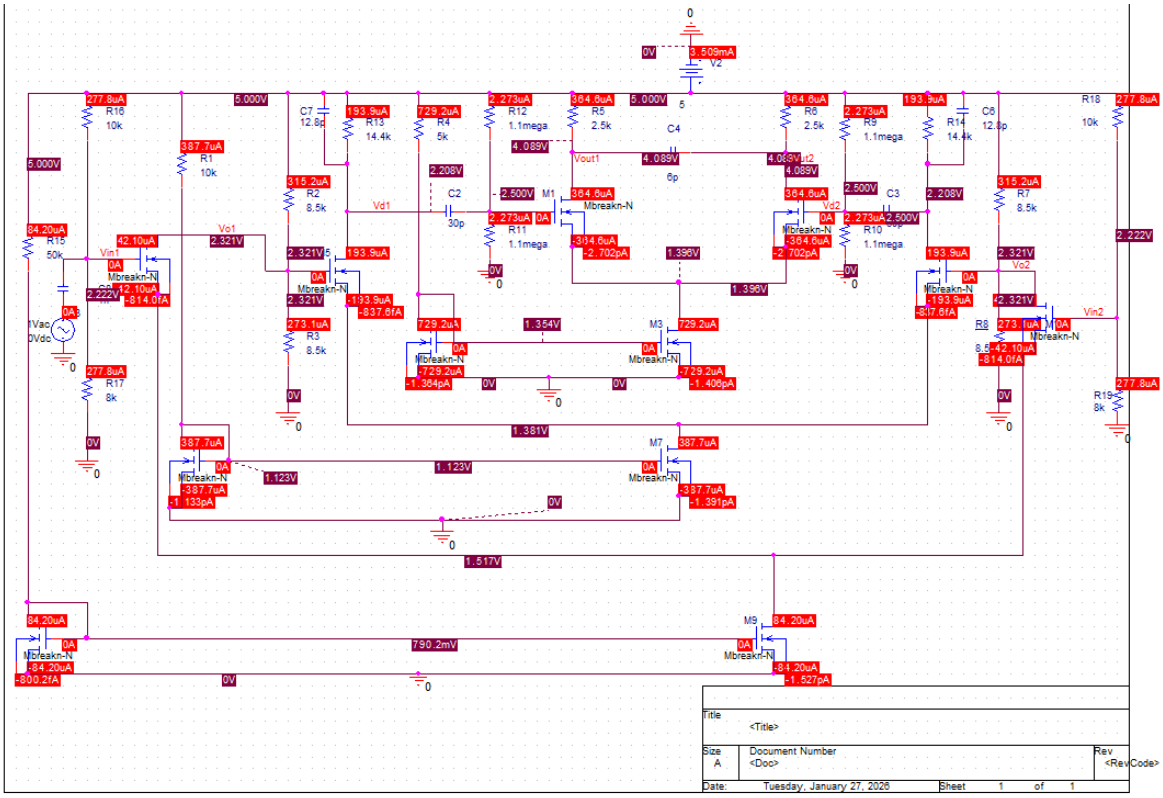

Figure 1: Bias Point Simulation: Schematic annotated with DC voltages and currents.

3 Small Signal Analysis

Parameters: Given $k = 1\frac{mA}{V^2}$.

$$g_{m11} = g_{m12} = 2\sqrt{kI_{D12}} = 410\,\mu S$$
$$g_{m5} = g_{m6} = 2\sqrt{kI_{D5}} = 881\,\mu S$$
$$g_{m1} = g_{m2} = 2\sqrt{kI_{D1}} = 1.21\,mS$$

Gain Calculations: Assuming $\lambda = 0$.

- **First Stage (A_1):**
$$A_1 = g_{m12} \cdot (R_2 \parallel R_3) = 1.74$$
- **Second Stage (A_2):**
$$A_2 = g_{m5} \cdot (R_{13} \parallel R_{12} \parallel R_{11}) = 12.36$$
- **Third Stage (A_3):**
$$A_3 = g_{m1} \cdot R_5 = 3$$

Total Gain:

$$A_p = A_1 \cdot A_2 \cdot A_3 = 64.5\,V/V$$
$$A_{p,dB} = 20\log_{10}(64.5) = 36.2\,dB$$

Error calculation vs Target (36dB):

$$err = \left(\frac{64.5}{63.1} - 1\right) \cdot 100\% = 2.2\%$$

3.1 Gain and Slope Verification

The AC sweep simulation confirms not only the gain but also the specified slopes for low and high frequency roll-off. As seen in Figure 2, the peak gain is $36.171\,dB$, which is extremely close to the calculated $36.2\,dB$.

Furthermore, we can verify the slopes using the cursor data points provided in the simulation:

1. Low Frequency Slope (M_1): Using the cursors at approximately $10\,Hz$ and $100\,Hz$ (one decade apart):

$$P_1 : (10.139\,Hz, -23.095\,dB)$$

$$P_2 : (100.231\,Hz, -3.195\,dB)$$

$$Slope_{low} = \frac{-3.195 - (-23.095)}{1\,decade} \approx +\mathbf{20\,dB/dec}$$

2. High Frequency Slope (M_2): Using the cursors at approximately $100\,MHz$ and $1\,GHz$ (one decade apart):

$$P_3 : (100.231\,MHz, -30.302\,dB)$$

$$P_4 : (997.7\,MHz, -70.250\,dB)$$

$$Slope_{high} = \frac{-70.250 - (-30.302)}{1\,decade} \approx -\mathbf{40\,dB/dec}$$

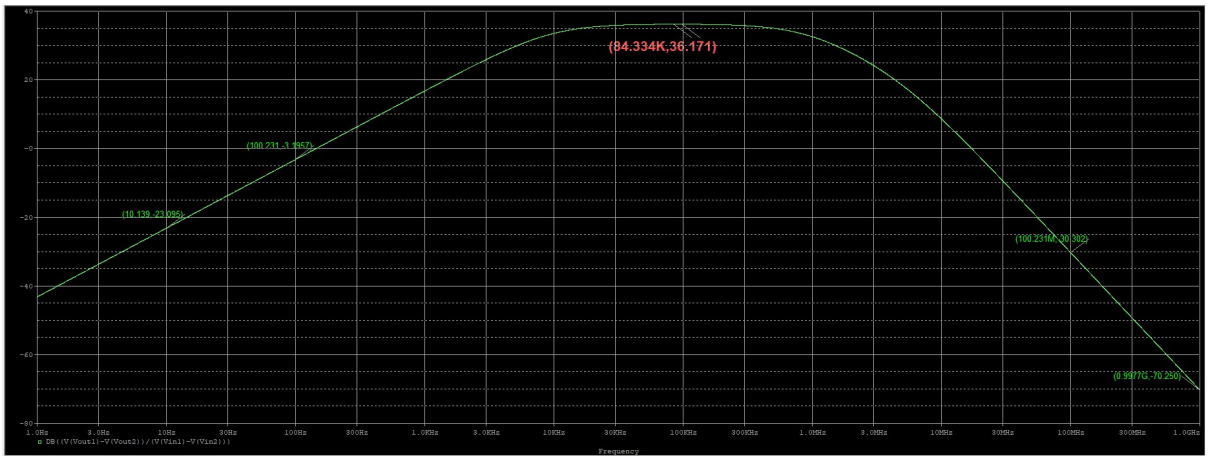

Figure 2: AC Sweep Simulation: The cursor indicates a peak gain of $36.171\,dB$.

4 Output Resistance & Frequency

Output Resistance (R_{out}):

$$R_{out} = 2R'_{out} = 2 \cdot R_5 = 5\,k\Omega$$

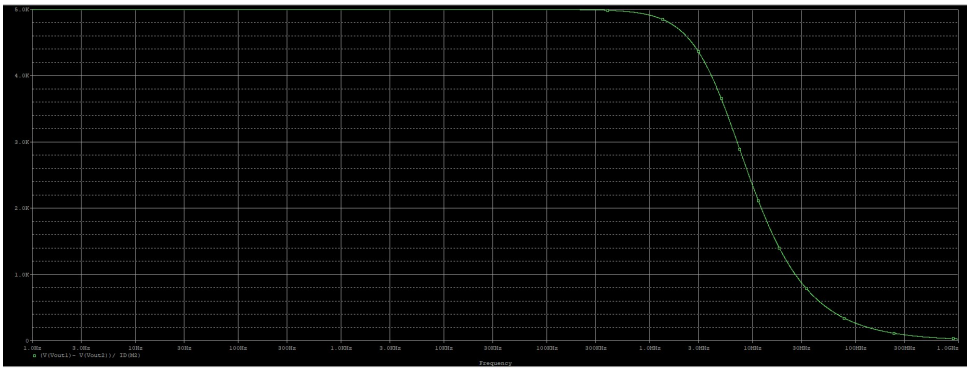

Figure 3: Schematic highlighting the output resistance (R_{out}).

Frequency Calculations: Since the internal parasitic capacitances of the MOSFETs are negligible compared to the schematic components, we use the explicit capacitors (C_2, C_7) to determine the cut-off frequencies.

4.1 Lower Cutoff Frequency (f_L)

The lower cutoff is dominated by the coupling capacitor C_2 between the second and third stage.

- **Capacitor:** $C_2 = 30\,pF$.
- **Resistance (R_{eq}):** The capacitor sees the bias resistors R_{11} and R_{12} in parallel (ignoring the negligible source impedance of the previous stage).

$$R_{eq} = R_{11} \parallel R_{12} = 1.1M\Omega \parallel 1.1M\Omega = 550\,k\Omega$$

- **Calculation:**

$$f_L = \frac{1}{2\pi \cdot R_{eq} \cdot C_2} = \frac{1}{2\pi \cdot 550k \cdot 30p} \approx 9.65\,kHz$$

Result: This closely matches the simulation result of $9.22\,kHz$ (vs Target $9\,kHz$).

4.2 Upper Cutoff Frequency (f_H)

The upper cutoff frequency is limited by the input network at the gate of M_1. The dominant pole is formed by the series resistance R_{13} and the gate capacitor C_7.
Parameters:

- Resistance: $R_{13} = 14.4\,k\Omega$
- Capacitance: $C_7 = 12.8\,pF$

Time Constant (τ_{in}):

$$\tau_{in} = R_{13} \cdot C_7 = 14.4k \cdot 12.8p \approx 184.3\,ns$$

Calculation:

$$f_H \approx \frac{1}{2\pi\tau_{in}} = \frac{1}{2\pi(184.3n)} \approx 863\,kHz$$

Result: This matches the simulation result of $879\,kHz$ with high accuracy (vs Target $900\,kHz$).

4.3 Frequency Verification

Figure 4 provides the simulation measurements for the 3dB cutoff frequencies, confirming the accuracy of the updated calculations.

	Evaluate	Measurement	Value
▶	☑	Cutoff_Highpass_3dB((((V(Vout1)-V...	9.22267k
	☑	Cutoff_Lowpass_3dB((((V(Vout1)-V...	879.07129k

Figure 4: Simulation Results: Cutoff_Highpass (f_L) and Cutoff_Lowpass (f_H) measurements.